

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12417948
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Lu; Jian-Hong et al.

Hybrid film scheme for self-aligned contact

Abstract

A method of forming a semiconductor device includes: forming a fin protruding above a substrate; forming a metal gate over the fin, the metal gate being surround by a dielectric layer; etching the metal gate to reduce a height of the metal gate, where after the etching, a recess is formed over the metal gate between gate spacers of the metal gate; lining sidewalls and a bottom of the recess with a semiconductor material; filling the recess by forming a dielectric material over the semiconductor material; forming a mask layer over the metal gate, where a first opening of the mask layer is directly over a portion of the dielectric layer adjacent to the metal gate; removing the portion of the dielectric layer to form a second opening in the dielectric layer, the second opening exposing an underlying source/drain region; and filling the second opening with a conductive material.

Inventors: Lu; Jian-Hong (Hsinchu, TW), Ho; Tsai-Jung (Xihu Township, TW), Hsieh; Bor Chiuan (Taoyuan, TW), Shih; Po-Cheng (Hsinchu, TW), Lee; Tze-Liang (Hsinchu, TW)

Applicant: Taiwan Semiconductor Manufacturing Co., Ltd. (Hsinchu, TW)

Family ID: 1000008820987

Assignee: Taiwan Semiconductor Manufacturing Co., Ltd. (Hsinchu, TW)

Appl. No.: 17/524830

Filed: November 12, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220246473 A1	Aug. 04, 2022

Related U.S. Application Data

us-provisional-application US 63145622 20210204

Publication Classification

Int. Cl.: **H10D64/68** (20250101); **H01L21/768** (20060101); **H10D30/01** (20250101); **H10D64/01** (20250101); **H10D84/01** (20250101); **H10D84/03** (20250101); H10D30/62 (20250101); H10D64/23 (20250101)

U.S. Cl.:

CPC **H01L21/76897** (20130101); **H10D30/024** (20250101); **H10D64/01** (20250101); **H10D84/0149** (20250101); **H10D84/0158** (20250101); **H10D84/038** (20250101); H10D30/6219 (20250101); H10D64/259 (20250101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9105490	12/2014	Wang et al.	N/A	N/A
9236267	12/2015	De et al.	N/A	N/A
9236300	12/2015	Liaw	N/A	N/A
9406804	12/2015	Huang et al.	N/A	N/A
9443769	12/2015	Wang et al.	N/A	N/A
9520482	12/2015	Chang et al.	N/A	N/A
9548366	12/2016	Ho et al.	N/A	N/A
9576814	12/2016	Wu et al.	N/A	N/A
9831183	12/2016	Lin et al.	N/A	N/A
9859386	12/2017	Ho et al.	N/A	N/A
10157783	12/2017	Chang et al.	N/A	N/A
10164053	12/2017	Chung et al.	N/A	N/A
10734233	12/2019	Zang et al.	N/A	N/A
2012/0139061	12/2011	Ramachandran	438/586	H01L 29/78
2015/0021683	12/2014	Xie	438/270	H10D 64/015
2020/0105577	12/2019	Liang et al.	N/A	N/A
2020/0251390	12/2019	More et al.	N/A	N/A
2021/0384316	12/2020	Yu et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102020119275	12/2020	DE	N/A
201732899	12/2016	TW	N/A
201810532	12/2017	TW	N/A
201913748	12/2018	TW	N/A
201937608	12/2018	TW	N/A

Primary Examiner: Gauthier; Steven B

Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application claims the benefit of the U.S. Provisional Application No. 63/145,622, filed on Feb. 4, 2021 and entitled “Hybrid Film SAC by Si Liner and SiN to Improve MD HM Bite/Chopping,” which application is hereby incorporated herein by reference.

BACKGROUND

(1) The semiconductor industry has experienced rapid growth due to continuous improvements in the integration density of a variety of electronic components (e.g., transistors, diodes, resistors, capacitors, etc.). For the most part, this improvement in integration density has come from repeated reductions in minimum feature size, which allows more components to be integrated into a given area.

(2) Fin Field-Effect Transistor (FinFET) devices are becoming commonly used in integrated circuits. FinFET devices have a three-dimensional structure that comprises a semiconductor fin protruding from a substrate. A gate structure, configured to control the flow of charge carriers within a conductive channel of the FinFET device, wraps around the semiconductor fin. For example, in a tri-gate FinFET device, the gate structure wraps around three sides of the semiconductor fin, thereby forming conductive channels on three sides of the semiconductor fin.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a perspective view of a Fin Field-Effect Transistor (FinFET) device, in accordance with some embodiments.

(3) FIGS. 2-6, 7A-7C, and 8-15 illustrate various cross-sectional views of a FinFET device at various stages of fabrication, in accordance with an embodiment.

(4) FIG. 16 illustrates a flow chart of method of forming a semiconductor device, in accordance with some embodiments.

DETAILED DESCRIPTION

(5) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact.

(6) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are

intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Throughout the discussion herein, unless otherwise specified, the same reference numeral in different figures refers to the same or similar component formed by a same or similar method using a same or similar material(s).

(7) Embodiments of the present disclosure are discussed in the context of forming a FinFET device, and in particular, in the context of forming self-aligned contact plugs for a FinFET device.

Although the disclosed embodiment is discussed using FinFET device as an example, the disclosed methods may also be used in other types of devices, such as planar devices.

(8) In some embodiments, when forming a self-aligned source/drain contact, a hybrid film structure is formed over a metal gate structure between gate spacers of the metal gate structure to reduce chopping effect. For example, the metal gate structure, which is surrounded by an inter-layer dielectric (ILD) layer, is recessed to form a recess between the gate spacers that extend along sidewalls of the metal gate structure. A semiconductor liner (e.g., a silicon liner) is conformally formed along sidewalls and a bottom of the recess, and a dielectric material (e.g., silicon nitride) is then formed on the semiconductor liner in the recess to fill the recess. The semiconductor liner and the dielectric material in the recess are collectively referred to as a hybrid film structure. A mask layer is then formed over the ILD layer, where an opening in the mask layer is directly over the hybrid film structure and a portion of the ILD layer disposed over a source/drain region. An etching process is performed next using the mask layer as an etching mask to remove the portion of the ILD layer to form an opening in the ILD layer, where the opening exposes the underlying source/drain region. Next, a conductive material is formed in the opening to form the self-aligned source/drain contact. Due to the etching selectivity provided by the hybrid film structure during the etching process to form the opening, chopping effect is reduced, which in turn reduces device defects related to short-circuit caused by the chopping effect.

(9) FIG. 1 illustrates an example of a FinFET 30 in a perspective view. The FinFET 30 includes a substrate 50 and a fin 64 protruding above the substrate 50. Isolation regions 62 are formed on opposing sides of the fin 64, with the fin 64 protruding above the isolation regions 62. A gate dielectric 66 is along sidewalls and over a top surface of the fin 64, and a gate electrode 68 is over the gate dielectric 66. Source/drain regions 80 are in the fin 64 and on opposing sides of the gate dielectric 66 and the gate electrode 68. FIG. 1 further illustrates reference cross-sections that are used in later figures. Cross-section B-B extends along a longitudinal axis of the gate electrode 68 of the FinFET 30. Cross-section A-A is perpendicular to cross-section B-B and is along a longitudinal axis of the fin 64 and in a direction of, e.g., a current flow between the source/drain regions 80. Cross-section C-C is parallel to cross-section B-B and is across the source/drain region 80. Subsequent figures refer to these reference cross-sections for clarity.

(10) FIGS. 2-6, 7A-7C, and 8-15 are cross-sectional views of a FinFET device 100 at various stages of fabrication, in accordance with an embodiment. The FinFET device 100 is similar to the FinFET 30 in FIG. 1, but with multiple fins and multiple gate structures. FIGS. 2-5 illustrate cross-sectional views of the FinFET device 100 along cross-section B-B. FIGS. 6, 7A, and 8-15 illustrate cross-sectional views of the FinFET device 100 along cross-section A-A. FIGS. 7B and 7C illustrate embodiment cross-sectional views of the FinFET device 100 along cross-section C-C.

(11) FIG. 2 illustrates a cross-sectional view of the substrate 50. The substrate 50 may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The substrate 50 may be a wafer, such as a silicon wafer. Generally, an SOI substrate includes a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon substrate or a glass substrate. Other substrates, such as a multi-layered

or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate **50** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof.

(12) Referring to FIG. 3, the substrate **50** shown in FIG. 2 is patterned using, for example, photolithography and etching techniques. For example, a mask layer, such as a pad oxide layer **52** and an overlying pad nitride layer **56**, is formed over the substrate **50**. The pad oxide layer **52** may be a thin film comprising silicon oxide formed, for example, using a thermal oxidation process. The pad oxide layer **52** may act as an adhesion layer between the substrate **50** and the overlying pad nitride layer **56**. In some embodiments, the pad nitride layer **56** is formed of silicon nitride, silicon oxynitride, silicon carbonitride, the like, or a combination thereof, and may be formed using low-pressure chemical vapor deposition (LPCVD) or plasma enhanced chemical vapor deposition (PECVD), as examples.

(13) The mask layer may be patterned using photolithography techniques. Generally, photolithography techniques utilize a photoresist material (not shown) that is deposited, irradiated (exposed), and developed to remove a portion of the photoresist material. The remaining photoresist material protects the underlying material, such as the mask layer in this example, from subsequent processing steps, such as etching. In this example, the photoresist material is used to pattern the pad oxide layer **52** and pad nitride layer **56** to form a patterned mask **58**, as illustrated in FIG. 3.

(14) The patterned mask **58** is subsequently used to pattern exposed portions of the substrate **50** to form trenches **61**, thereby defining semiconductor fins **64** (e.g., **64A** and **64B**) between adjacent trenches **61** as illustrated in FIG. 3. In some embodiments, the semiconductor fins **64** are formed by etching trenches in the substrate **50** using, for example, reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etching process may be anisotropic. In some embodiments, the trenches **61** may be strips (viewed from in the top) parallel to each other, and closely spaced with respect to each other. In some embodiments, the trenches **61** may be continuous and surround the semiconductor fins **64**. The semiconductor fins **64** may also be referred to as fins **64** hereinafter.

(15) The fins **64** may be patterned by any suitable method. For example, the fins **64** may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers, or mandrels, may then be used to pattern the fins.

(16) FIG. 4 illustrates the formation of an insulation material between neighboring semiconductor fins **64** to form isolation regions **62**. The insulation material may be an oxide, such as silicon oxide, a nitride, the like, or a combination thereof, and may be formed by a high density plasma chemical vapor deposition (HDP-CVD), a flowable CVD (FCVD) (e.g., a CVD-based material deposition in a remote plasma system and post curing to make it convert to another material, such as an oxide), the like, or a combination thereof. Other insulation materials and/or other formation processes may be used. In the illustrated embodiment, the insulation material is silicon oxide formed by a FCVD process. An anneal process may be performed once the insulation material is formed. A planarization process, such as a chemical mechanical polish (CMP), may remove any excess insulation material and form top surfaces of the isolation regions **62** and top surfaces of the semiconductor fins **64** that are coplanar (not shown). The patterned mask **58** (see FIG. 3) may also

be removed by the planarization process.

(17) In some embodiments, the isolation regions **62** include a liner, e.g., a liner oxide (not shown), at the interface between the isolation region **62** and the substrate **50**/semiconductor fins **64**. In some embodiments, the liner oxide is formed to reduce crystalline defects at the interface between the substrate **50** and the isolation region **62**. Similarly, the liner oxide may also be used to reduce crystalline defects at the interface between the semiconductor fins **64** and the isolation region **62**. The liner oxide (e.g., silicon oxide) may be a thermal oxide formed through a thermal oxidation of a surface layer of substrate **50**, although other suitable method may also be used to form the liner oxide.

(18) Next, the isolation regions **62** are recessed to form shallow trench isolation (STI) regions **62**. The isolation regions **62** are recessed such that the upper portions of the semiconductor fins **64** protrude from between neighboring STI regions **62**. The top surfaces of the STI regions **62** may have a flat surface (as illustrated), a convex surface, a concave surface (such as dishing), or a combination thereof. The top surfaces of the STI regions **62** may be formed flat, convex, and/or concave by an appropriate etch. The isolation regions **62** may be recessed using an acceptable etching process, such as one that is selective to the material of the isolation regions **62**. For example, a dry etch, or a wet etch using dilute hydrofluoric (dHF) acid, may be performed to recess the isolation regions **62**.

(19) FIGS. **2** through **4** illustrate an embodiment of forming fins **64**, but fins may be formed in various different processes. For example, a top portion of the substrate **50** may be replaced by a suitable material, such as an epitaxial material suitable for an intended type (e.g., N-type or P-type) of semiconductor devices to be formed. Thereafter, the substrate **50**, with epitaxial material on top, is patterned to form semiconductor fins **64** that comprise the epitaxial material.

(20) As another example, a dielectric layer can be formed over a top surface of a substrate; trenches can be etched through the dielectric layer; homoepitaxial structures can be epitaxially grown in the trenches; and the dielectric layer can be recessed such that the homoepitaxial structures protrude from the dielectric layer to form fins.

(21) In yet another example, a dielectric layer can be formed over a top surface of a substrate; trenches can be etched through the dielectric layer; heteroepitaxial structures can be epitaxially grown in the trenches using a material different from the substrate; and the dielectric layer can be recessed such that the heteroepitaxial structures protrude from the dielectric layer to form fins.

(22) In embodiments where epitaxial material(s) or epitaxial structures (e.g., the heteroepitaxial structures or the homoepitaxial structures) are grown, the grown material(s) or structures may be in situ doped during growth, which may obviate prior and subsequent implantations although in situ and implantation doping may be used together. Still further, it may be advantageous to epitaxially grow a material in an NMOS region different from the material in a PMOS region. In various embodiments, the fins **64** may comprise silicon germanium ($\text{Si}_{\text{sub}.x}\text{Ge}_{\text{sub}.1-x}$, where x can be between 0 and 1), silicon carbide, pure or substantially pure germanium, a III-V compound semiconductor, a II-VI compound semiconductor, or the like. For example, the available materials for forming III-V compound semiconductor include, but are not limited to, InAs, AlAs, GaAs, InP, GaN, InGaAs, InAlAs, GaSb, AlSb, AlP, GaP, and the like.

(23) FIG. **5** illustrates the formation of dummy gate structure **75** over the semiconductor fins **64**. Dummy gate structure **75** includes gate dielectric **66** and gate electrode **68**, in some embodiments. A mask **70** may be formed over the dummy gate structure **75**. To form the dummy gate structure **75**, a dielectric layer is formed on the semiconductor fins **64**. The dielectric layer may be, for example, silicon oxide, silicon nitride, multilayers thereof, or the like, and may be deposited or thermally grown.

(24) A gate layer is formed over the dielectric layer, and a mask layer is formed over the gate layer. The gate layer may be deposited over the dielectric layer and then planarized, such as by a CMP. The mask layer may be deposited over the gate layer. The gate layer may be formed of, for

example, polysilicon, although other materials may also be used. The mask layer may be formed of, for example, silicon nitride or the like.

(25) After the layers (e.g., the dielectric layer, the gate layer, and the mask layer) are formed, the mask layer may be patterned using acceptable photolithography and etching techniques to form mask **70**. The pattern of the mask **70** then may be transferred to the gate layer and the dielectric layer by an acceptable etching technique to form gate electrode **68** and gate dielectric **66**, respectively. The gate electrode **68** and the gate dielectric **66** cover respective channel regions of the semiconductor fins **64**. The gate electrode **68** may also have a lengthwise direction substantially perpendicular to the lengthwise direction of respective semiconductor fins **64**.

(26) The gate dielectric **66** is shown to be formed over the fins **64** (e.g., over top surfaces and sidewalls of the fins **64**) and over the STI regions **62** in the example of FIG. 5. In other embodiments, the gate dielectric **66** may be formed by, e.g., thermal oxidization of a material of the fins **64**, and therefore, may be formed over the fins **64** but not over the STI regions **62**. These and other variations are fully intended to be included within the scope of the present disclosure.

(27) FIGS. 6, 7A, and 8-15 illustrate the cross-sectional views of further processing of the FinFET device **100** along cross-section A-A (along a longitudinal axis of the fin **64**). Note that in FIGS. 6, 7A, and 8, three dummy gate structures **75** (e.g., 75A, 75B, and 75C) are formed over the fin **64**. One skilled in the art will appreciate that more or less than three gate structures may be formed over the fin **64**, these and other variations are fully intended to be included within the scope of the present disclosure.

(28) As illustrated in FIG. 6, lightly doped drain (LDD) regions **65** are formed in the fins **64**. The LDD regions **65** may be formed by a plasma doping process. The plasma doping process may include forming and patterning masks such as a photoresist to cover the regions of the FinFET that are to be protected from the plasma doping process. The plasma doping process may implant N-type or P-type impurities in the fins **64** to form the LDD regions **65**. For example, P-type impurities, such as boron, may be implanted in the fin **64** to form the LDD regions **65** for a P-type device. As another example, N-type impurities, such as phosphorus, may be implanted in the fin **64** to form the LDD regions **65** for an N-type device. In some embodiments, the LDD regions **65** abut the channel region of the FinFET device **100**. Portions of the LDD regions **65** may extend under gate electrode **68** and into the channel region of the FinFET device **100**. FIG. 6 illustrates a non-limiting example of the LDD regions **65**. Other configurations, shapes, and formation methods of the LDD regions **65** are also possible and are fully intended to be included within the scope of the present disclosure. For example, LDD regions **65** may be formed after gate spacers **87** are formed. In some embodiments, the LDD regions **65** are omitted. For simplicity, the LDD regions **65** are not illustrated in subsequent figures, with the understanding the LDD regions **65** may be formed in the fin **64**.

(29) Still referring to FIG. 6, after the LDD regions **65** are formed, gate spacers **87** are formed around the dummy gate structures **75**. The gate spacer **87** may include a first gate spacer **72** and a second gate spacer **86**. For example, the first gate spacer **72** may be a gate seal spacer and is formed on opposing sidewalls of the gate electrode **68** and on opposing sidewalls of the gate dielectric **66**. The second gate spacer **86** is formed on the first gate spacer **72**. The first gate spacer **72** may be formed of a nitride, such as silicon nitride, silicon oxynitride, silicon carbonitride, the like, or a combination thereof, and may be formed using, e.g., a thermal oxidation, CVD, or other suitable deposition process. The second gate spacer **86** may be formed of silicon nitride, silicon carbonitride, a combination thereof, or the like using a suitable deposition method. In an embodiment, the first gate spacer **72** is formed of silicon nitride, and an atomic ratio between silicon and nitrogen (e.g., a ratio between the atomic percentages of silicon and nitrogen) is between about 0.7 and about 1.3.

(30) In an embodiment, the gate spacer **87** is formed by first conformally depositing a first gate spacer layer over the FinFET device **100**, then conformally depositing a second gate spacer layer

over the deposited first gate spacer layer. Next, an anisotropic etch process, such as a dry etch process, is performed to remove a first portion of the second gate spacer layer disposed on upper surfaces of the FinFET device **100** (e.g., the upper surface of the mask **70**) while keeping a second portion of the second gate spacer layer disposed along sidewalls of the gate structures. The second portion of the second gate spacer layer remaining after the anisotropic etch process forms the second gate spacer **86**. The anisotropic etch process also removes a portion of the first gate spacer layer disposed outside of the sidewalls of the second gate spacer **86**, and the remaining portion of the first gate spacer layer forms the first gate spacer **72**.

(31) The shapes and formation methods of the gate spacer **87** as illustrated in FIG. **6** are merely non-limiting examples, and other shapes and formation methods are possible. These and other variations are fully intended to be included within the scope of the present disclosure.

(32) Next, in FIG. **7A**, recesses are formed in the fins **64** adjacent to the dummy gate structures **75**, e.g., between adjacent dummy gate structures **75** and/or next to a dummy gate structure **75**. Source/drain regions **80** are then formed in the recesses. The recesses are formed by, e.g., an anisotropic etching process using the dummy gate structures **75** and the gate spacers **87** as an etching mask, in some embodiments, although any other suitable etching process may also be used.

(33) Next, as illustrated in FIG. **7A**, the source/drain regions **80** are formed in the recesses. The source/drain regions **80** are formed by epitaxially growing a material in the recesses, using suitable methods such as metal-organic CVD (MOCVD), molecular beam epitaxy (MBE), liquid phase epitaxy (LPE), vapor phase epitaxy (VPE), selective epitaxial growth (SEG), the like, or a combination thereof.

(34) As illustrated in FIG. **7A**, the epitaxial source/drain regions **80** may have surfaces raised from respective surfaces of the fins **64** (e.g. raised above the non-recessed upper surface **64U** of the fins **64**) and may have facets. The source/drain regions **80** of the adjacent fins **64** may merge to form a continuous epitaxial source/drain region **80** (see FIG. **7B**). In some embodiments, the source/drain regions **80** of the adjacent fins **64** do not merge together and remain separate source/drain regions **80** (see FIG. **7C**). In some embodiments, the resulting FinFET is an n-type FinFET, and source/drain regions **80** comprise silicon carbide (SiC), silicon phosphorous (SiP), phosphorous-doped silicon carbon (SiCP), or the like. In some embodiments, the resulting FinFET is a p-type FinFET, and source/drain regions **80** comprise SiGe, and a p-type impurity such as boron or indium.

(35) The epitaxial source/drain regions **80** may be implanted with dopants to form source/drain regions **80** followed by an anneal process. The implanting process may include forming and patterning masks such as a photoresist to cover the regions of the FinFET device **100** that are to be protected from the implanting process. The source/drain regions **80** may have an impurity (e.g., dopant) concentration in a range from about $1\text{E}19\text{ cm.sup.-3}$ to about $1\text{E}21\text{ cm.sup.-3}$. P-type impurities, such as boron or indium, may be implanted in the source/drain region **80** of a P-type transistor. N-type impurities, such as phosphorous or arsenide, may be implanted in the source/drain regions **80** of an N-type transistor. In some embodiments, the epitaxial source/drain regions may be in situ doped during growth.

(36) Next, as illustrated in FIG. **8**, a contact etch stop layer (CESL) **89** is formed over the structure illustrated in FIG. **7A**. The CESL **89** functions as an etch stop layer in a subsequent etching process, and may comprise a suitable material such as silicon oxide, silicon nitride, silicon oxynitride, combinations thereof, or the like, and may be formed by a suitable formation method such as CVD, PVD, combinations thereof, or the like.

(37) Next, a first interlayer dielectric (ILD) **90** is formed over the CESL **89** and over the dummy gate structures **75** (e.g., **75A**, **75B**, and **75C**). In some embodiments, the first ILD **90** is formed of a dielectric material such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate Glass (BPSG), undoped silicate glass (USG), or the like, and may be deposited by any suitable method, such as CVD, PECVD, or FCVD. A planarization process, such

as CMP, may be performed to remove the mask **70** and to remove portions of the CESL **89** disposed over the gate electrode **68**. After the planarization process, the top surface of the first ILD **90** is level with the top surface of the gate electrode **68**.

(38) Next, in FIG. **9**, a gate-last process (sometimes referred to as replacement gate process) is performed to replace the gate electrode **68** and the gate dielectric **66** with an active gate (may also be referred to as a replacement gate or a metal gate) and active gate dielectric material(s), respectively. Therefore, the gate electrode **68** and the gate dielectric **66** may be referred to as dummy gate electrode and dummy gate dielectric, respectively, in a gate-last process.

(39) Referring to FIG. **9**, the dummy gate structures **75A**, **75B**, and **75C** (see FIG. **8**) are replaced by replacement gate structures **97A**, **97B**, and **97C**, respectively. The replacement gate structures **97** (e.g., **97A**, **97B**, and **97C**) may also be referred to as metal gate structures. In accordance with some embodiments, to form the replacement gate structures **97** (e.g., **97A**, **97B**, or **97C**), the gate electrode **68** and the gate dielectric **66** directly under the gate electrode **68** are removed in an etching step(s), so that recesses (not shown) are formed between the gate spacers **87**. Each recess exposes the channel region of a respective fin **64**. During the dummy gate removal, the gate dielectric **66** may be used as an etch stop layer when the gate electrode **68** is etched. The gate dielectric **66** may then be removed after the removal of the gate electrode **68**.

(40) Next, a gate dielectric layer **94**, a barrier layer **96**, a work function layer **98**, and a gate electrode **99** are formed in the recesses for the replacement gate structures **97**. The gate dielectric layer **94** is deposited conformally in the recesses, such as on the top surfaces and the sidewalls of the fins **64** and on sidewalls of the gate spacers **87**, and on a top surface of the first ILD **90** (not shown). In accordance with some embodiments, the gate dielectric layer **94** comprises silicon oxide, silicon nitride, or multilayers thereof. In other embodiments, the gate dielectric layer **94** includes a high-k dielectric material, and in these embodiments, the gate dielectric layers **94** may have a k value greater than about 7.0, and may include a metal oxide or a silicate of Hf, Al, Zr, La, Mg, Ba, Ti, Pb, and combinations thereof. The formation methods of gate dielectric layer **94** may include molecular beam deposition (MBD), atomic layer deposition (ALD), PECVD, and the like.

(41) Next, the barrier layer **96** is formed conformally over the gate dielectric layer **94**. The barrier layer **96** may comprise an electrically conductive material such as titanium nitride, although other materials, such as tantalum nitride, titanium, tantalum, or the like, may alternatively be utilized. The barrier layer **96** may be formed using a CVD process, such as PECVD. However, other alternative processes, such as sputtering, metal organic chemical vapor deposition (MOCVD), or ALD, may alternatively be used.

(42) Next, the work function layer **98**, such as a P-type work function layer or an N-type work function layer, may be formed in the recesses over the barrier layers **96** and before the gate electrode **99** is formed, in some embodiments. Exemplary P-type work function metals that may be included in the gate structures for P-type devices include TiN, TaN, Ru, Mo, Al, WN, ZrSi.sub.2, MoSi.sub.2, TaSi.sub.2, NiSi.sub.2, WN, other suitable P-type work function materials, or combinations thereof. Exemplary N-type work function metals that may be included in the gate structures for N-type devices include Ti, Ag, TaAl, TaAlC, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, other suitable N-type work function materials, or combinations thereof. A work function value is associated with the material composition of the work function layer, and thus, the material of the work function layer is chosen to tune its work function value so that a target threshold voltage V_t is achieved in the device that is to be formed. The work function layer(s) may be deposited by CVD, physical vapor deposition (PVD), and/or other suitable process.

(43) Next, a seed layer (not shown) is formed conformally over the work function layer **98**. The seed layer may include copper, titanium, tantalum, titanium nitride, tantalum nitride, the like, or a combination thereof, and may be deposited by ALD, sputtering, PVD, or the like. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sublayers formed of different materials. For example, the seed layer

comprises a titanium layer and a copper layer over the titanium layer.

(44) Next, the gate electrode **99** is deposited over the seed layer, and fills the remaining portions of the recesses. The gate electrode **99** may be made of a metal-containing material such as Cu, Al, W, the like, combinations thereof, or multi-layers thereof, and may be formed by, e.g., electroplating, electroless plating, or other suitable method. After the formation of the gate electrode **99**, a planarization process, such as a CMP, may be performed to remove the excess portions of the gate dielectric layer **94**, the barrier layer **96**, the work function layer **98**, the seed layer, and the gate electrode **99**, which excess portions are over the top surface of the first ILD **90**. The resulting remaining portions of the gate dielectric layer **94**, the barrier layer **96**, the work function layer **98**, the seed layer, and the gate electrode **99** thus form the replacement gate structure **97** of the resulting FinFET device **100**.

(45) Referring next to FIG. **10**, an etch stop layer **105** is formed (e.g., selectively) on the upper surface of the first ILD **90**. The etch stop layer **105** may be a suitable dielectric material, such as silicon nitride or silicon oxynitride formed on the upper surface of the first ILD **90** by, e.g., CVD, ALD, combinations thereof, or the like. For example, a patterned mask layer may be formed to cover the replacement gate structures **97** and the gate spacers **87** while exposing the first ILD **90**, and the etch stop layer **105** is then formed over the exposed first ILD **90**. After the etch stop layer **105** is formed, the patterned mask layer is removed. As another example, an upper layer of the first ILD **90** may be converted into the etch stop layer **105**, e.g., by a nitridation process using a nitrogen-containing gas or a nitrogen-containing plasma.

(46) Next, the replacement gate structures **97** are recessed to form recesses **103** between respective gate spacers **87**. In some embodiments, a metal gate etch-back process is performed to remove upper portions of the replacement gate structures **97**, and as a result, recesses **103** are formed between the gate spacers **87** after the metal gate etch-back process. A suitable etching process, such as dry etch, wet etch, or combinations thereof, may be performed as the metal gate etch-back process.

(47) Still referring to FIG. **10**, after the recesses **103** are formed, a capping layer **101** is formed on the recessed replacement gate structures **97**. In some embodiments, the capping layer **101** is formed of an electrically conductive material, such as tungsten, using a suitable deposition method, such as ALD, CVD, PVD, or the like. In the example of FIG. **10**, the capping layer **101** extends continuously along the upper surface of the replacement gate structure **97** from an inner sidewall of a gate spacer **87** to an opposing inner sidewall of a respective gate spacer **87**. In other words, the capping layer **101** completely covers the upper surface of the replacement gate structures **97**, in the illustrated embodiment. In some embodiments, the capping layer **101** is omitted.

(48) Next, in FIG. **11**, the recesses **103** are filled by a hybrid film **108**. As illustrated in FIG. **11**, a semiconductor material **107** (also referred to as a semiconductor liner) is formed conformally along sidewalls and bottoms of the recesses **103**, and along the upper surface of the etch stop layer **105**. Next, a dielectric material **109** is formed over the semiconductor material **107** to fill the recesses **103**. The dielectric material **109** may also be formed outside the recesses **103** over portions of the semiconductor material **107** between replacement gate structures **97**. The semiconductor material **107** and the dielectric material **109** are collectively referred to as a hybrid film **108**.

(49) In an example embodiment, the semiconductor material **107** is amorphous silicon (a-Si) formed by a suitable deposition process such as LPCVD, PECVD, ALD, or the like, using a silicon-containing process gas such as SiH₄, Si₂H₆, SiH₂Cl₂, combinations thereof, or the like. A flow rate of the process gas may be between about 50 standard cubic centimeters per minute (sccm) and about 1000 sccm. A pressure of the deposition process may be between about 0.1 torr and about 10 torr, and a temperature of the deposition process may be between about 300° C. and about 500° C. A thickness T of the semiconductor material **107** is between about 0.3 nm and about 5 nm (e.g., 0.3 nm < T < 5 nm), in an embodiment. An atomic percentage (at %) of hydrogen in the semiconductor material **107** is less than about 15 at % (e.g.,

H<15 at %), in an embodiment. In the discussion herein, the conformal semiconductor material **107** may also be referred to as a silicon liner, with the understanding that other suitable material may also be used as the semiconductor material **107**.

(50) In an example embodiment, the dielectric material **109** is silicon nitride formed by a suitable formation method, such as ALD, PECVD, LPCVD, or the like. A atomic ratio between silicon and nitrogen (e.g., Si:N atomic ratio) in the dielectric material **109** is between about 0.7 and about 1.3. Besides silicon nitride (e.g., SiN), other suitable dielectric material, such as silicon carbon nitride (SiCN), silicon carbon oxynitride (SiCON), and silicon carbon oxide (SiCO), may also be used as the dielectric material **109**. In some embodiments, the dielectric material **109** is chosen to be different from the material(s) of the gate spacers **87** to provide etching selectivity in subsequent etching processes, such as the etching process to form openings **121** (see FIG. **13**).

(51) The as-deposited semiconductor material **107** may (or may not) include some crystalline structures such as polycrystalline structures, which include grains therein. Some grains may be connected to each other, while some other grains may be buried in amorphous structures. The copper in, e.g., the gate electrode **99** may diffuse upward along the grain boundaries to overlying layers that will be formed subsequently. Therefore, to reduce the upward diffusion of copper, after the deposition of the semiconductor material **107**, a treatment process (also referred to as an amorphization process) is performed to convert the polycrystalline structures (if any) in the semiconductor material **107** into amorphous structures, so that the entire semiconductor material **107** is amorphous. Since amorphous semiconductor material **107** does not have grain boundaries, it has better ability to prevent copper from diffusing through. The treatment process is performed after the semiconductor material **107** is formed and before the dielectric material **109** is formed, in some embodiments.

(52) In accordance with some embodiments, the treatment process (e.g., a plasma process) for the as-deposited semiconductor material **107** is performed using a process gas comprising NH₃, N₂, or combination thereof. Other gases, such as argon, may also be added. In the treatment process, the semiconductor material **107** is bombarded. The bombardment destroys the crystalline structures. In addition, with hydrogen and nitrogen atoms being in the process gas (e.g., NH₃), hydrogen and nitrogen may be added into the semiconductor material **107**. Accordingly, the semiconductor material **107** may comprise hydrogen doped therein as a result of the treatment process. In accordance with some embodiments, after the amorphization process, the semiconductor material **107** has a hydrogen atomic percentage in the range between about 0.5 at % and about 10 at %.

(53) In accordance with some embodiments, the treatment process is or comprises a plasma treatment, which may be a direct plasma process, with the plasma being generated in the same process chamber where the FinFET device **100** is treated. The plasma treatment process is performed using both a High-Frequency Radio-Frequency (HFRF) power (e.g., with a frequency about 13.56 MHz) and a Low-Frequency Radio-Frequency (LFRF) power (e.g., with a frequency of about 350 KHz). The HFRF power is used for ionization and to generate plasma, and the LFRF power is used for bombarding the semiconductor material **107** for amorphization purpose. In accordance with some embodiments of the present disclosure, the HFRF power is in the range between about 50 watts and about 2000 watts.

(54) The LFRF power is selected to be in a suitable range. If the LFRF power is too low (e.g., lower than about 90 watts), the amorphization process may not be effective. If the LFRF power is too high (e.g., higher than about 135 watts), there may be severe plasma-induced damage to layer/structures underlying the semiconductor material **107**. In accordance with some embodiments of the present disclosure, the LFRF power is selected to be in the range between about 90 watts and about 135 watts to avoid the aforementioned issues.

(55) Next, in FIG. **12**, a planarization process, such as CMP, is performed to remove excess portions of the semiconductor material **107** and excess portions of the dielectric material **109**, which excess portions are over the upper surface of the first ILD **90**. As illustrated in FIG. **12**, the

planarization process also remove the etch stop layer **105** and top portions of the gate spacers **87**. After the planarization process, the remaining portions of the semiconductor material **107** and the dielectric material **109** disposed between (remaining portions of) respective gate spacers **87** are referred to as hybrid film structures **108**.

(56) Next, a second ILD **111** is formed over the first ILD **90**, the gate spacers **87**, and the hybrid film structures **108**. The second ILD **111** may be formed of a same or similar material using a same or similar formation method as the first ILD **90**, thus details are not repeated. The second ILD **111** may serve as an etch stop layer in a subsequent etching process.

(57) Next, an etch stop layer stack **112** is formed over the second ILD **111** and patterned to have an opening **120** in the etch stop layer stack **112**. In some embodiments, the etch stop layer stack **112** includes a plurality of layers (also referred to as sublayers), where each of the plurality of layers is formed of a different material. In the example of FIG. **12**, the etch stop layer stack **112** includes a first layer **113**, a second layer **115**, and a third layer **117** formed successively over the second ILD **111**. The first layer **113** is formed of tungsten doped carbide (WDC, also known as tungsten doped silicon carbide), the second layer **115** is formed of an oxide (e.g., silicon oxide), and the third layer **117** is formed of silicon (e.g. Si). The different materials of the sublayers of the etch stop layer stack **112** combine to provide a target level of etching selectivity for, e.g., the subsequent etching process to form openings **121** in FIG. **13**.

(58) Next, the etch stop layer stack **112** is patterned to form an opening **120** in the etch stop layer stack **112**. For example, a patterned mask layer **119** (e.g., a patterned photoresist layer) is formed over the etch stop layer stack **112**, and an anisotropic etching process is then performed using the patterned mask layer **119** as an etching mask to pattern the patterned mask layer **119**. The number and the location of the opening **120** in FIG. **12** is merely a non-limiting example. One skilled in the art will readily appreciate that other numbers of openings **120** may be formed at other locations. In some embodiments, the anisotropic etching process includes a plurality of etching steps, where each of the etching steps uses a different etchant to selectively remove a sublayer of the etch stop layer stack **112**. The anisotropic etching process may stop when the second ILD **111** is exposed. The patterned mask layer **119** (e.g., a patterned photoresist layer) may be removed after the opening **120** is formed, e.g., by an ashing process.

(59) Next, in FIG. **13**, an etching process is performed to remove portions of the second ILD **111** and portions of the first ILD **90** underlying the opening **120**. After the etching process, openings **121** are formed in the first ILD **90** that expose the underlying source/drain regions **80**. A depth D1 of the opening **121** may be between about 20 nm and about 80 nm, as an example. The etching process may be any suitable etching process, such as a dry etch (e.g., a plasma etching process), a wet etch, combinations thereof, or the like. The etching process may use an etchant that is selective to (e.g., having a higher etching rate for) the material (e.g., oxide) of the first ILD **90** and the second ILD **111**, such that the first ILD **90** and the second ILD **111** are removed without substantially attacking other layers of the FinFET device **100**. For example, an etching processing using C.sub.xF.sub.y, H.sub.2, Ar, combinations thereof, or the like may be performed to remove the exposed portions of the first ILD **90** and the second ILD **111**. In some embodiments, after the etching process to remove the exposed portions of the first ILD **90** and the second ILD **111**, another etching process using an etchant selective to the material of the CESL **89** is performed to remove the CESL **89** exposed by the openings **121** and to expose the source/drain regions **80**. Note that the portions of the first ILD **90**/second ILD **111** underlying the opening **120** are completely removed in preparation for forming self-aligned contacts (SAC) **123** (see FIG. **14**).

(60) In some embodiments, the etching process to form the openings **121** also removes top portions of the gate spacers **87** and top portions of the hybrid film structure **108** underlying the opening **120** in FIG. **12**. However, the hybrid film structure **108**, with its unique structure, dimension (e.g., 0.3 nm<T<5 nm) and material composition (e.g., atomic percentage of H<15 at %, and the atomic ratio of Si:N between 0.7 and 1.3), provides good etching selectivity in the etching process to form the

openings **121**, such that the hybrid film structure **108** is etched at a slower rate than the gate spacers **87**. In other words, the etching process to form the opening **121** recesses the hybrid film structure **108** less than the gate spacers **87**. In the example of FIG. **13**, after the etching process is finished, the upper surface **87U** of the gate spacer **87** is sloped (e.g. slanted). For example, there is an offset **D2** between an uppermost edge of the upper surface **87U** (which may be level with the upper surface of the hybrid film structure **108**) and a lowermost edge of the upper surface **87U**. The offset **D2** may be, e.g., 10 nm or less. The unintended removal (e.g., etching) of the hybrid film structure **108** and the gate spacer **87** is referred to as chopping effect.

(61) The current disclosure embodiment, by using the hybrid film structure **108**, among other features, reduces the chopping effect significantly, which in turn reduces product defect related with the chopping effect. To appreciate the advantage of the present disclosure, consider a reference design where the hybrid film structure **108** is replaced by a single layer of dielectric material (e.g., silicon nitride). In other words, in the reference design, a single silicon nitride layer is formed over the replacement gate structure **97** and fills the space between respective gate spacers **87**. The chopping effect would remove significant portions of the single silicon nitride layer and the gate spacers **87**. In subsequent processing to fill the openings **121** with a conductive material to form self-aligned contacts **123**, the conductive material may fill spaces left by the removed portions of the single silicon nitride layer and the gate spacers **87**, which may result in short circuit (e.g., unintended electrical connection) between the replacement gate structure **97** and the source/drain region **80**, or between the source/drain regions **80** on opposing sides of the replacement gate structure **97**. The current disclosure avoids or reduces the short-circuit defects caused by the chopping effect. The current disclosed structure and method have been implemented and tested in production, and measurements have confirmed that chopping effect induced short-circuit defects have been reduced by four orders of magnitude.

(62) Next, in FIG. **14**, a conductive material is formed in the openings **121** to fill the openings **121**. The conductive material may be copper, cobalt, tungsten, combinations thereof, or the like, and may be formed by a suitable formation method such as PVD, CVD, PECVD, ALD, or the like. Next, a planarization process, such as CMP, is performed to remove the etch stop layer stack **112** and the second ILD **111**. The remaining portions of the conductive material in the openings **121** form self-aligned contacts **123**. As illustrated in FIG. **14**, after the planarization process, the first ILD **90**, the hybrid film structure **108**, the gate spacers **87**, and the self-aligned contacts **123** have a coplanar upper surface.

(63) Next, in FIG. **15**, an interconnect structure **148** is formed over the first ILD **90** to interconnect the underlying electrical components (e.g., the FinFETs) to form functional circuits. The interconnect structure **148** includes a plurality of dielectric layers (e.g., **125**, **129**, **133**) and conductive features (e.g., **141**, **143**, **145**, **147**) formed in the plurality of dielectric layers. Note that in the description herein, unless otherwise specified, conductive features and conductive materials refer to electrically conductive features and electrical conductive materials, respectively.

(64) The dielectric layers **125/129/133** may also be referred to as inter-metal dielectric (IMD) layers. The IMD layers **125/129/133** may be formed of one or more suitable dielectric materials such as silicon oxide, silicon nitride, low-k dielectrics such as carbon doped oxides, extremely low-k dielectrics such as porous carbon doped silicon dioxide, combinations of these, or the like. The IMD layers **125/129/133** may be formed through a process such as chemical vapor deposition (CVD), although any suitable process may be utilized. Etch stop layers **127** and **131** may be formed between adjacent IMD layers. The etch stop layers **127** and **131** may be formed of silicon nitride using PECVD, although other dielectric materials such as nitride, carbide, boride, combinations thereof, or the like, and alternative techniques of forming the etch stop layers, such as LPCVD, PVD, or the like, could alternatively be used.

(65) In FIG. **15**, gate contacts **141** are formed to extend through the IMD layer **125**, the dielectric material **109**, and the semiconductor material **107** to electrically couple to the replacement gate

structures **97**, e.g., through the capping layer **101**. In the example of FIG. **15**, the lower surface of the gate contact **141** physically contacts and extends along the upper surface of the underlying capping layer **101**. In addition, source/drain contacts **143** are formed to extend through the IMD layer **125** to electrically couple to the self-aligned contacts **123**. FIG. **15** further illustrates conductive lines **145** and vias **147** formed in IMD layers **129** and **133**, respectively. The gate contacts **141**, source/drain contacts **143**, conductive lines **145**, and via **147** are formed of one or more conductive materials (e.g., copper, tungsten, cobalt), and may be formed using any suitable method, such as damascene, dual-damascene, or the like. Note that the number and the location of the conductive features in the interconnect structure **148** are for illustration purpose only and not limiting. Other conductive features at other locations may be formed. In addition, some conductive features formed may not be in the cross-section of FIG. **15**, thus are not visible (e.g., shown) in FIG. **15**.

(66) FIG. **16** illustrates a flow chart of a method **1000** of forming a semiconductor device, in accordance with some embodiments. It should be understood that the embodiment method shown in FIG. **16** is merely an example of many possible embodiment methods. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. For example, various steps as illustrated in FIG. **16** may be added, removed, replaced, rearranged and repeated.

(67) Referring to FIG. **16**, at block **1010**, a metal gate structure is formed over a semiconductor fin, wherein the metal gate structure is surrounded by an interlayer dielectric (ILD) layer. At block **1020**, the metal gate structure is recessed below an upper surface of gate spacers that extend along opposing sidewalls of the metal gate structure, wherein after the recessing, a recess is formed between the gate spacers. At block **1030**, sidewalls and a bottom of the recess are lined with a semiconductor material. At block **1040**, after the lining, the recess is filled with a dielectric material. At block **1050**, a patterned etch stop layer stack is formed over the dielectric material, the semiconductor material, and the ILD layer, wherein a first opening in the patterned etch stop layer stack is directly over the dielectric material, the semiconductor material, and a first portion of the ILD layer adjacent to the gate spacers. At block **1060**, a second opening is formed in the ILD layer by etching through the first opening to remove the first portion of the ILD layer, wherein the second opening exposes a source/drain region adjacent to the metal gate structure. At block **1070**, the second opening is filled with a conductive material.

(68) Embodiments may achieve advantages. For example, the disclosed structure and method reduces chopping effect, thereby preventing or reducing device defect due to short-circuit caused by the chopping effect. Measurements have confirmed that chopping effect induced defect rate has been reduced by four orders of magnitude in production by using the disclosed structure and method.

(69) In accordance with an embodiment, a method of forming a semiconductor device includes: forming a metal gate structure over a semiconductor fin, wherein the metal gate structure is surrounded by an interlayer dielectric (ILD) layer; recessing the metal gate structure below an upper surface of gate spacers that extend along opposing sidewalls of the metal gate structure, wherein after the recessing, a recess is formed between the gate spacers; lining sidewalls and a bottom of the recess with a semiconductor material; after the lining, filling the recess with a dielectric material; forming a patterned etch stop layer stack over the dielectric material, the semiconductor material, and the ILD layer, wherein a first opening in the patterned etch stop layer stack is directly over the dielectric material, the semiconductor material, and a first portion of the ILD layer adjacent to the gate spacers; forming a second opening in the ILD layer by etching through the first opening to remove the first portion of the ILD layer, wherein the second opening exposes a source/drain region adjacent to the metal gate structure; and filling the second opening with a conductive material. In an embodiment, the semiconductor material is amorphous silicon. In an embodiment, the dielectric material is silicon nitride. In an embodiment, the method further includes, before recessing the metal gate structure, selectively forming an etch stop layer over an

upper surface of the ILD layer distal from the semiconductor fin, wherein a first portion of the semiconductor material contacts and extends along the etch stop layer. In an embodiment, the method further includes, after filling the recess and before forming the patterned etch stop layer stack, performing a planarization process, wherein the planarization process removes the first portion of the semiconductor material and the etch stop layer. In an embodiment, the method further includes, after the recessing and before the lining, forming a capping layer over the metal gate structure, wherein the capping layer physically contacts the metal gate structure and the semiconductor material. In an embodiment, forming the second opening comprises performing an etching process to remove the first portion of the ILD layer, wherein the etching process recesses the gate spacers more than the dielectric material and the semiconductor material. In an embodiment, forming the patterned etch stop layer stack comprises: forming an etch stop layer stack by forming a first sublayer, a second sublayer, and a third sublayer successively over the ILD layer, wherein the first sublayer, the second sublayer, and the third sublayer comprise different materials; forming a patterned mask layer over the etch stop layer stack; and patterning the etch stop layer stack using the patterned mask layer as an etching mask. In an embodiment, the first sublayer is formed of tungsten doped silicon carbide, the second sublayer is formed of silicon oxide, and the third sublayer is formed of silicon. In an embodiment, the method further includes, after filling the second opening: removing the patterned etch stop layer stack; forming an inter-metal dielectric (IMD) layer over the ILD layer; and forming a first via and a second via that extend through the IMD layer, wherein the first via further extends through the dielectric material and the semiconductor material and is electrically coupled to the metal gate structure, wherein the second via is electrically coupled to the conductive material.

(70) In accordance with an embodiment, a method of forming a semiconductor device includes: forming a first metal gate structure over a fin; forming a second metal gate structure over the fin and adjacent to the first metal gate structure, wherein the first metal gate structure and the second metal gate structure are surrounded by an interlayer dielectric (ILD) layer; recessing the first metal gate structure and the second metal gate structure to form a first opening between first gate spacers and to form a second opening between second gate spacers, respectively, wherein the first gate spacers are on opposing sidewalls of the first metal gate structure, and the second gate spacers are on opposing sidewalls of the second metal gate structure; conformally forming a semiconductor material along sidewalls and bottoms of the first opening and the second opening; filling the first opening and the second opening by forming a dielectric material on the semiconductor material; and forming a self-aligned contact between the first metal gate structure and the second metal gate structure, wherein the self-aligned contact is electrically coupled to an underlying source/drain region between the first metal gate structure and the second metal gate structure. In an embodiment, forming the self-aligned contact comprises, after filling the first opening and the second opening: forming an etch stop layer stack over the ILD layer, wherein the etch stop layer stack comprises a plurality of layers, each of the plurality of layers being formed of a different material; forming a first opening in the etch stop layer stack over the first metal gate structure and over a first portion of the ILD layer between the first metal gate structure and the second metal gate structure; removing the first portion of the ILD layer to form a second opening in the ILD layer, wherein the second opening exposes the underlying source/drain region; and filling the second opening with a conductive material, wherein the conductive material extends continuously from the first metal gate structure to the second metal gate structure. In an embodiment, forming the etch stop layer stack comprises: forming a first layer of tungsten doped silicon carbide over the ILD layer; forming a second layer of silicon oxide over the first layer; and forming a third layer of silicon over the second layer. In an embodiment, the semiconductor material is amorphous silicon, and the dielectric material is silicon nitride. In an embodiment, the method further includes, after the recessing and before conformally forming the semiconductor material, forming a conductive capping layer on the first metal gate structure and on the second metal gate structure. In an

embodiment, the conductive material contacts and extends along an upper surface of the conductive capping layer distal from the fin.

(71) In accordance with an embodiment, a method of forming a semiconductor device includes: forming a fin protruding above a substrate; forming a metal gate over the fin, wherein the metal gate is surrounded by a dielectric layer; etching the metal gate to reduce a height of the metal gate, wherein after the etching, a recess is formed over the metal gate between gate spacers of the metal gate; lining sidewalls and a bottom of the recess with a semiconductor material; filling the recess by forming a dielectric material over the semiconductor material; forming a patterned mask layer over the metal gate and the dielectric layer, wherein a first opening of the patterned mask layer is directly over a portion of the dielectric layer adjacent to the metal gate; removing the portion of the dielectric layer to form a second opening in the dielectric layer, wherein the second opening exposes an underlying source/drain region; and filling the second opening with a conductive material. In an embodiment, the semiconductor material is amorphous silicon, and the dielectric material is silicon nitride. In an embodiment, the method further comprising, after etching the metal gate and before the lining, selectively forming a capping layer on the metal gate. In an embodiment, the second opening exposes a sidewall of the gate spacers.

(72) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method of forming a semiconductor device, the method comprising: forming a metal gate structure over a semiconductor fin, wherein the metal gate structure is surrounded by an interlayer dielectric (ILD) layer; recessing the metal gate structure below an upper surface of gate spacers that extend along opposing sidewalls of the metal gate structure, wherein after the recessing, a recess is formed between the gate spacers; lining sidewalls and a bottom of the recess with a semiconductor material; after the lining, filling the recess with a dielectric material; forming a patterned etch stop layer stack over the dielectric material, the semiconductor material, and the ILD layer, wherein a first opening in the patterned etch stop layer stack is directly over the dielectric material, the semiconductor material, and a first portion of the ILD layer adjacent to the gate spacers; forming a second opening in the ILD layer by etching through the first opening to remove the first portion of the ILD layer, wherein forming the second opening comprises performing an etching process to remove the first portion of the ILD layer, wherein the etching process recesses the gate spacers more than the dielectric material and the semiconductor material, wherein the second opening exposes a source/drain region adjacent to the metal gate structure; and filling the second opening with a conductive material.
2. The method of claim 1, wherein the semiconductor material is amorphous silicon.
3. The method of claim 2, wherein the dielectric material is silicon nitride.
4. The method of claim 1, further comprising, before recessing the metal gate structure, selectively forming an etch stop layer over an upper surface of the ILD layer distal from the semiconductor fin, wherein a first portion of the semiconductor material contacts and extends along the etch stop layer.
5. The method of claim 4, further comprising, after filling the recess and before forming the patterned etch stop layer stack, performing a planarization process, wherein the planarization process removes the first portion of the semiconductor material and the etch stop layer.

6. The method of claim 4, wherein lining the sidewalls and the bottom of the recess comprises forming a conformal layer of the semiconductor material in the recess and along the upper surface of the ILD layer.
7. The method of claim 1, further comprising, after the recessing and before the lining, forming a capping layer over the metal gate structure, wherein the capping layer physically contacts the metal gate structure and the semiconductor material.
8. The method of claim 1, wherein forming the patterned etch stop layer stack comprises: forming an etch stop layer stack by forming a first sublayer, a second sublayer, and a third sublayer successively over the ILD layer, wherein the first sublayer, the second sublayer, and the third sublayer comprise different materials; forming a patterned mask layer over the etch stop layer stack; and patterning the etch stop layer stack using the patterned mask layer as an etching mask.
9. The method of claim 8, wherein the first sublayer is formed of tungsten doped silicon carbide, the second sublayer is formed of silicon oxide, and the third sublayer is formed of silicon.
10. The method of claim 1, further comprising, after filling the second opening: removing the patterned etch stop layer stack; forming an inter-metal dielectric (IMD) layer over the ILD layer; and forming a first via and a second via that extend through the IMD layer, wherein the first via further extends through the dielectric material and the semiconductor material and is electrically coupled to the metal gate structure, wherein the second via is electrically coupled to the conductive material.
11. A method of forming a semiconductor device, the method comprising: forming a first metal gate structure over a fin; forming a second metal gate structure over the fin and adjacent to the first metal gate structure, wherein the first metal gate structure and the second metal gate structure are surrounded by an interlayer dielectric (ILD) layer; recessing the first metal gate structure and the second metal gate structure to form a first opening between first gate spacers and to form a second opening between second gate spacers, respectively, wherein the first gate spacers are on opposing sidewalls of the first metal gate structure, and the second gate spacers are on opposing sidewalls of the second metal gate structure; after the recessing, forming a conductive capping layer on the first metal gate structure and on the second metal gate structure; after forming the conductive capping layer, conformally forming a semiconductor material along sidewalls and bottoms of the first opening and the second opening; filling the first opening and the second opening by forming a dielectric material on the semiconductor material; and forming a self-aligned contact between the first metal gate structure and the second metal gate structure, wherein the self-aligned contact is electrically coupled to an underlying source/drain region between the first metal gate structure and the second metal gate structure.
12. The method of claim 11, wherein forming the self-aligned contact comprises, after filling the first opening and the second opening: forming an etch stop layer stack over the ILD layer, wherein the etch stop layer stack comprises a plurality of layers, each of the plurality of layers being formed of a different material; forming a first opening in the etch stop layer stack over the first metal gate structure and over a first portion of the ILD layer between the first metal gate structure and the second metal gate structure; removing the first portion of the ILD layer to form a second opening in the ILD layer, wherein the second opening exposes the underlying source/drain region; and filling the second opening with a conductive material, wherein the conductive material extends continuously from the first metal gate structure to the second metal gate structure.
13. The method of claim 12, wherein forming the etch stop layer stack comprises: forming a first layer of tungsten doped silicon carbide over the ILD layer; forming a second layer of silicon oxide over the first layer; and forming a third layer of silicon over the second layer.
14. The method of claim 11, wherein the semiconductor material is amorphous silicon, and the dielectric material is silicon nitride.
15. A method of forming a semiconductor device, the method comprising: forming a fin protruding above a substrate; forming a metal gate over the fin, wherein the metal gate is surrounded by a

dielectric layer; etching the metal gate to reduce a height of the metal gate, wherein after the etching, a recess is formed over the metal gate between gate spacers of the metal gate; after etching the metal gate, forming a conductive capping layer in the recess on the metal gate; after forming the conductive capping layer, lining sidewalls and a bottom of the recess with a semiconductor material; filling the recess by forming a dielectric material over the semiconductor material; forming a patterned mask layer over the metal gate and the dielectric layer, wherein a first opening of the patterned mask layer is directly over a portion of the dielectric layer adjacent to the metal gate; removing the portion of the dielectric layer to form a second opening in the dielectric layer, wherein the second opening exposes an underlying source/drain region; and filling the second opening with a conductive material.

16. The method of claim 15, wherein the semiconductor material is amorphous silicon, and the dielectric material is silicon nitride.

17. The method of claim 15, wherein the second opening exposes a sidewall of the gate spacers.

18. The method of claim 15, further comprising, before etching the metal gate, selectively forming an etch stop layer over an upper surface of the dielectric layer distal from the fin.

19. The method of claim 18, wherein lining the sidewalls and the bottom of the recess comprises conformally forming the semiconductor material in the recess and along the upper surface of the dielectric layer, wherein a first portion of the semiconductor material along the upper surface of the dielectric layer contacts and extends along the etch stop layer.

20. The method of claim 19, further comprising, after filling the recess and before forming the patterned mask layer, performing a planarization process, wherein the planarization process removes the first portion of the semiconductor material and the etch stop layer.
